

Linear regression assumptions

Every model made by some assumptions, if that assumptions are not validate

The model performance become less

Linear regression assumptions is divided into two ways

- *Based on features or input columns*
- *Based on Errors or Residuals*

Assumptions based input columns:

- *Linearity:*
 - *There is a linear relationship between dependent and independent variable*
 - *Model name is Linear regression*
 - *it is an important property to maintain linear relationship*

How to check:

- *Scatter plots are the best to identify the linear relationship*
- *we cant draw more than 3D data*

We want Linearity

- *Multicollinearity:*
 - *Input variables should be independent each other*
 - *If input columns having relation then this multicollinearity*
 - *Multicollinearity is similar to correlation*
 - *Any model: It is better to have input columns No Multicollinearity*

We don't want Multicollinearity

- *How to check*

Math analogy

x_1	x_2	$y = b_0 + b_1 * x_1 + b_2 * x_2$
2	4	
4	8	
6	12	
8	16	

x_1 and x_2 both are dependent : $x_2 = 2 * x_1$

$$y = b_o + b_1 * x_1 + b_2 * 2x_1$$

$$y = b_o + x_1[b_1 + 2b_2]$$

$$b_1 + 2b_2 = 0$$

maths says, if you want find two unknowns we need two equations

but above we have one equation and two unknowns : Maths fail

VIF: Variance inflation factor

VIF will give which variable cause the multicollineraity

$$VIF = \frac{1}{1-R_{square}}$$

Procedure:

we already know what is R_{square}

R_{square} : Explanibility

How much input variable explaining about variance in output variable

Assume that we have 3 input columns are there: x_1, x_2, x_3

$$Y = b_o + b_1 * x_1 + b_2 * x_2 + b_3 * x_3$$

Main goal : Idenrify the Multicollineraity between input columns

For that we need to calculate VIF value

VIF value depends on R_{square} value

Step – 1: Consider x_1 as output variable ,

x_2 and x_3 consider as input

$$x_1 = b_0 + b_2 * x_2 + b_3 * x_3$$

the above equation will give R_{square}

R_{square} will give VIF value

$$\text{assume: } R_{square} = 0.9 \quad VIF = \frac{1}{1-0.9} = \frac{1}{0.1} = 10$$

Step – 2: Consider x_2 as output variable ,

x_1 and x_3 consider as input

$$x_2 = b_0 + b_1 * x_1 + b_3 * x_3$$

the above equation will give R_{square}

R_{square} will give VIF value

$$R_{square} = 0.95 \quad VIF = \frac{1}{1-0.95} = \frac{1}{0.05} = 20$$

Step – 3: Consider x_3 as output variable ,

x_1 and x_2 consider as input

$$x_3 = b_0 + b_1 * x_1 + b_2 * x_2$$

the above equation will give R_{square}

$$R_{square} = 0.80 \quad VIF = \frac{1}{1-0.8} = \frac{1}{0.2} = 5$$

Conclusion:

In industry $VIF > 10$ we can consider as Multicollinearity is there

x_2 gives $VIF = 20$,so x_2 has more multicollinearity with x_1 and x_3

we will not consider x_2 for model development

- 1) Remove the variables
- 2) PCA: Principle component analysis

So Multicollinearity concept also used as Feature selection

Till now the Feature selection methods are

- 1) *p – value*
- 2) *Adj R_Square*
- 3) *VIF*

Assumptions based on Errors or Residuals:

- *NO Autocorrelation*
Means errors are not correlated each other
- *Homoscedasticity:*
 - *The variace of errors is same across all independent varaiables*
 - *variance should be constant*
- *Errors Follows Normality*
The distribution of errors follows Normal distribution

NO Autocorrelation

Imagin that we have a data of 10 observations

each 10 observations we have 10 actual output

10 observations we pass to model , then model will give 10 prediction

we will get 10 errors

	X	y_a	y _p	ε
1		✓	✓	ε ₁
2				ε ₂
3				ε ₃
4				
10				ε ₁

Auto correlation:

Finding the relation between the values of same column

Errors should not have autocorrelation

How to check:

Durbin watson test(DW test)

$e_1, e_2, e_3, e_4, \dots \dots e_n$

Step – 1: Difference between the errors

$$d_1 = e_2 - e_1$$

$$d_2 = e_3 - e_2$$

$$d_3 = e_4 - e_3$$

$$d_t = e_{t+1} - e_t$$

Nr = summation of difference of errors squares

$$= d_1^2 + d_2^2 + d_3^2 + \dots + d_t^2 =$$

$$(e_2 - e_1)^2 + (e_3 - e_2)^2 + \dots + (e_{t+1} - e_t)^2$$

Dr = sum of square of errors

$$DW = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2}$$

x	y	y_p $= -2.6268x$ $+ 1,129.2$
10	1100	
20	1200	
35	985	
40	750	
50	1215	
45	1000	

x	y	$y_p = -2.6268x + 1,129.2$	$error = (y_a - y_p)$
10	1100	$-2.6268 * 10 + 1,129.2 = 1102.9$	-2.9
20	1200	$-2.6268 * 20 + 1,129.2 = 1076.7$	123.3
35	985	$-2.6268 * 35 + 1,129.2 = 1037.3$	-52.3
40	750	$-2.6268 * 40 + 1,129.2 = 1024.1$	-274.1
50	1215	$-2.6268 * 50 + 1,129.2 = 997.8$	217.1
45	1000	$-2.6268 * 45 + 1,129.2 = 1011$	-11

$DR = \text{sum of square errors}$

$$e_1^2 + e_2^2 + e_3^2 + e_4^2 + e_5^2 + e_6^2$$

$$(-2.9)^2 + (123.3)^2 + (-52.3)^2 + (-274.1)^2 + (217.1)^2 + (-11)^2 = 140,330.81$$

$Nr = \text{Difference of errors}$

$$d_1^2 = (e_2 - e_1)^2 = (126.2)^2$$

$$d_2^2 = (e_3 - e_2)^2 = (-175.6)^2$$

$$d_3^2 = (e_4 - e_3)^2 = (-221.9)^2$$

$$d_4^2 = (e_5 - e_4)^2 = (491.3)^2$$

$$d_5^2 = (e_6 - e_5)^2 = (-228.1)^2$$

$$(126.2)^2 + (-175.6)^2 + (-221.9)^2 + (491.3)^2 + (-228.1)^2 = 389,406.71$$

$$DW = \frac{\sum_{t=2}^n (e_t - e_{t-1})^2}{\sum_{t=1}^n e_t^2} = \frac{389,406.71}{140,330.81} = 2.77$$

Dw test interpretation:

Ranges 0 to 4

DW = 0 to 2 : Positive relation

DW = ~2 : No relation

DW = 2 to 4: Negative relation

Homoscedasticity :

- *Errors have constant variance throughout the data*
- *Constant variance means homoscedasticity*
- *The distribution of errors is constant*

Heteroscedasticity :

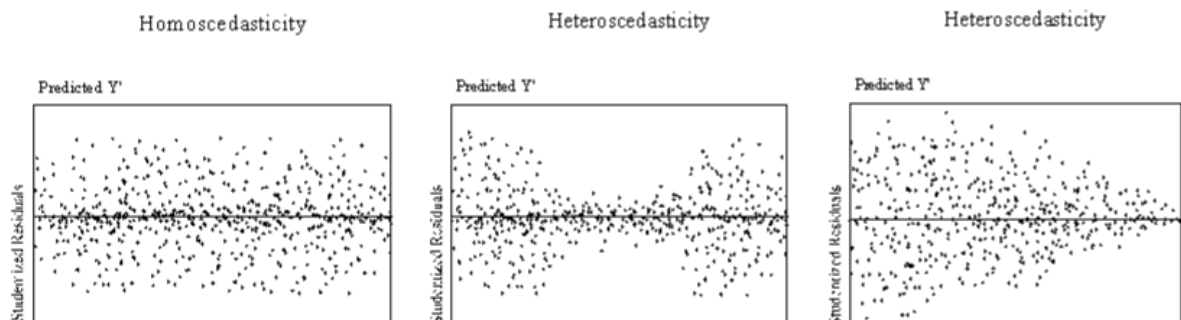
- *Errors have not constant variance(tunnel shape) throughout the data*
- *No Constant variance means heteroscedasticity*
- *The distribution of errors is not constant*

If errors distribution changes model will not perform all the levels of data

How to check:

Residuals vs Fitted plot

Errors vs Predictions



Normality:

Errors should follow Normal distribution

We will perform Hypothesis testing for Selection of variables

which means we will find the p value based on statistical test

we already know that Normal distributed data easy to perform the statistical tests

How to check:

Using Histogram

Q – Q plot : Quantile – Quantile plot



All together

1) *Linearity : we want Linearity between input and output*

Scatter plot

2) *No Multicollinearity : we don't want multicollinearity between input columns*

VIF

3) *No autocorrelation: We don't want relation between errors*

Dw test

4) *Homoscedasticity: we want constant variance of errors*

Residual vs fitted value or Errors vs Prediction

5) *Normality: we want error follows Normal distribution*

QQplot